

Computational design and reproducibility audit of candidate betaine responsive circuits in *Corynebacterium glutamicum*

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Abstract

We present a computational design workflow and an audited collection of DNA fragments for a proposed glycine betaine responsive circuit in *Corynebacterium glutamicum* ATCC 13032. The workflow combines accession reconciliation, protein-domain annotation, comparative sequence analysis, genomic operator-motif screening, promoter and translation-context design, and component-level sequence checks. The collection contains 97 candidate fragments derived from a 95-fragment baseline: 73 output-layer fragments, 18 PhoQ sensor candidates and six EnvZ sensor candidates. Accession reconciliation identified a 46-residue difference at the N terminus of the primary sensory protein. Rescoring an existing set of 304 upstream motif hits against an intergenic background left 82 at the selected empirical threshold. A fresh rerun of the frozen release reproduced 38 selected output files byte for byte and verified the documented regulator-label correction. Transcript-start analysis exposed dependence on the reporter junction and limitations of a benchmark anchored to known starts. Betaine sensing, signal transfer, transcriptional activation and biological utility remain experimentally untested. The contribution is a reproducible design case study linking candidate sequences and intended comparisons to explicit assumptions, provenance and computational limits.

Keywords: *Corynebacterium glutamicum*; synthetic biology; two-component signaling; candidate design; sequence provenance; reproducibility

1. Introduction

Combining bacterial sensory proteins, transcriptional regulators and expression elements requires decisions that are not resolved by sequence assembly alone. *Corynebacterium glutamicum* has characterized promoter systems and bicistronic expression designs [1,2], but their combination with a heterologous sensor requires compatibility across ligand recognition, kinase signaling, regulator-promoter interaction and translation initiation. Activity measured for an isolated part or a source construct does not establish that the assembled circuit will function.

The proposed circuit couples a betaine-recognizing chemoreceptor region to a histidine kinase and its cognate response regulator, which would control transcription from an engineered promoter. In *C. glutamicum* ATCC 13032, betaine supports

osmoprotection, and neither its synthesis nor degradation was detected in the published experiments [3]. This motivates examining an externally supplied input, although uptake and osmoprotection could affect both exposure and reporter behavior. Matilla et al. measured betaine binding to the isolated PacA (ECA_RS10935) ligand-binding domain from *Pectobacterium atrosepticum* ($K_D = 7.5 \pm 0.1 \mu\text{M}$; Table 1 and Figure 6 of ref. 4) [4]. McpX from *Sinorhizobium meliloti* provides a second betaine-binding candidate [5], and a deposited PacA structure contains bound betaine [6]. A PctC/PhoQ-based γ -aminobutyric acid sensor in *Escherichia coli* provides precedent for chimeric chemoreceptor–kinase signaling [7]. These sources support component selection rather than the function of the proposed fusions in *C. glutamicum*.

This study reconstructs the design as a traceable set of sequence-level choices and tests whether selected computational outputs can be regenerated. It examines protein-source discrepancies, sensitivity of motif screening to genomic background, promoter–translation junction assumptions and changes between successive candidate panels. Explicit assumptions and reproducible sequences provide a defined starting point for experimental evaluation without requiring reconstruction of the underlying design history. No engineered strain or functioning inducible system is reported.

2. Methods

2.1 Study inputs and version boundary

The source collection comprised protein and genome sequences, cached database records and literature supplements, design manifests, analysis outputs and Python scripts. The host genome was *C. glutamicum* ATCC 13032 accession BX927147.1. The consolidated order_manifest_final_v4.csv defined a 95-fragment baseline. A later revision, P1 (Proposal 1), added two operator-scramble controls to produce 97 records. Thirty-nine conditional sequence edits and two translation-initiation additions modified existing records without increasing their number. Identifiers retain links to baseline parents. All 97 records are included, regardless of historical optional or deferred labels.

The additional reproducibility audit used the complete source archive in release v0.6.0 (commit 73213731043b04bb85660bd8edb2f6f8d24e9ae1; DOI 10.5281/zenodo.23132170). Separately downloaded GitHub and Zenodo source ZIPs had identical SHA-256 hashes, and all 426 entries in the included file-hash manifest matched. The contemporaneous GitHub main branch was one commit ahead, with changes limited to README.md and CITATION.cff; it was not used for computation.

Published measurements, computations on the candidate sequences and proposed circuit behavior were treated as distinct evidence. Counts refer to DNA fragments, not complete plasmids or engineered strains. Sequence hashes and component intervals accompany the resource. Component maps use zero-based, half-open intervals; genomic scan coordinates are one-based and inclusive.

2.2 Source reconciliation and protein annotation

Protein-source accessions were reconciled using cached UniProt and RefSeq records. The primary sensory sequence was PacA, WP_043878053.1; the secondary sensor was McpX, Q92SH9. PctD, Q9HVF8, was annotated but not retained among the 24 sensor fragments. The primary signaling pair was *Salmonella* Typhimurium LT2 PhoQ/PhoP, P0DM80/P0DM78; the alternative was *E. coli* EnvZ/OmpR, P0AEJ4/P0AA16. The retired database accession P14147 was reconciled to the strain-consistent PhoQ record P0DM80.

Two HMMER procedures served different purposes [8]. Comparative protein screening extracted the best available receiver (REC) and dimerization and histidine phosphotransfer (DHp) profile hits using the archived extractor. Dedicated domain annotation applied Pfam gathering thresholds with hmmscan --cut_ga and a permissive search at $E = 10$ to inspect weak or absent assignments. Permissive hits were distinguished from accepted domain assignments. Transmembrane boundaries came from archived DeepTMHMM 1.0.57 outputs [9]; re-execution regenerated the annotation tables without repeating neural-network inference.

2.3 Comparative protein screening

The comparison set contained six donor or reference proteins—four response regulators and two histidine kinases—and 26 native proteins comprising 13 of each class. Full-length and extracted-domain comparisons used Biopython pairwise alignment [10], BLOSUM62, gap-open and extension scores of -11 and -1 , and unpenalized terminal gaps. Aligned-column identity was the number of identical residues divided by alignment columns after terminal-gap trimming. A second output used the shorter sequence length as denominator. Both definitions were retained; identity was used for descriptive screening rather than as a prediction of cross-phosphorylation.

2.4 Operator motifs and genomic background sensitivity

The cached RegulonDB input contained 36 confirmed *E. coli* PhoP site records, of which 29 had the modal length of 17 nucleotides and were used to construct the PhoP matrix [11]. Seven different-length records were excluded from this matrix. A separate OmpR matrix used 45 records and was retained for potential follow-on analysis of the alternative branch; no OmpR genome-scan result is reported here. Nine confirmed *Salmonella* PhoP sites from PRODORIC MX000339 provided a second PhoP source [12]. Position-weight matrices used a pseudocount of 0.5 per nucleotide per position and $\log_2$ odds relative to the selected genomic background.

The PhoP scan evaluated both genomic strands against a dinucleotide-shuffled genome generated with seed 1337. Empirical tail fractions were computed from null scores, and overlapping candidate windows were reduced to local peaks on each strand. The HIGH_CONCERN software label identified noncoding hits within 400 bases upstream of an annotated coding-sequence start, using gene orientation; it denoted a screening category. The main comparison used a tail-fraction threshold of

10^{-4} . Recovery of the *E. coli* sites used to build the matrix served as an implementation check.

For background sensitivity, annotated coding intervals were merged and the intervening segments identified. A new matrix used intergenic base composition; its null pooled both-strand scores from independently dinucleotide-shuffled intergenic segments. The original 304 flagged sites were rescored, without performing a new exhaustive enumeration under the revised model. Empirical tail fractions were not treated as binding probabilities or false-discovery rates. Historical Poisson calculations were excluded from biological inference because they did not reproduce the full window-merging and site-selection procedure.

2.5 Candidate construction and selection logic

The output layer combined sourced Psyn, PsodA and Ptuf promoter material with designed operator contexts [1]. The two operator sequences represented *Salmonella* PhoP sites associated with phoP and mgtA. Families varied operator position and identity, substitution versus insertion geometry, surrounding sequence, retained promoter context and translation elements. Backgrounds were generated by seeded rejection sampling; constraints and seeds are preserved in the scripts. Candidate selection emphasized interpretable comparisons and spacing coverage rather than a learned expression score.

The substitution series placed a 16-base operator at 14 positions, with gap values 0–13. Here, *gap* is the interval between the operator’s downstream edge and the selected core-promoter segment beginning at the assigned –35 element; it is not distance to the transcription start. Total fragment length and GC content remain fixed, while local sequence context varies. The insertion series adds intervening sequence, changing length as well as spacing. Three controls at gap 7 replace the substitution background or insertion spacer while retaining the operator. Translation candidates use a bicistronic arrangement—a short upstream coding sequence, or fore-cistron, coupled to downstream translation—derived from Liu et al. [2]. Eleven of twelve characterized fore-cistrons were retained after excluding one containing an incompatible NcoI site. Initiation and downstream translation-element mutations provide mechanistic comparisons. The larger source sequence census was not used as an expression-training dataset.

Eight sensor-junction families combined sensory regions with PhoQ or EnvZ signaling regions. Each family contained three local boundary offsets, –1, 0 and +1, giving 24 fragments. Families differed in retained transmembrane, linker and signaling regions. Donor residue coordinates, domain boundaries and encoded proteins were recorded. Synonymous redesign screened DNA against the shared 35-enzyme catalogue while preserving the intended protein sequence. Local heptad-register calculations assessed candidate seven-residue structural alignments as heuristics; low-information transmembrane assignments were not treated as functional evidence.

2.6 Transcript-start and translation-context assessment

Transcript-start analysis used cached supplementary tables from Pfeifer-Sancar et al. [13]. Of 2,591 source rows, 2,522 promoter contexts were mapped. A model combined the distance from an assigned -10 element to the transcription start site (TSS) with local nucleotide context. Normalized scores were used to compare alternative promoter–translation junction hypotheses.

The archived benchmark split records into halves with seed 1337. Each supplied context was already anchored to a known TSS, restricting candidate positions. Its success rates therefore do not measure unconstrained start-site prediction in a new construct. Normalized outputs were interpreted as heuristic weights, and junction revisions remained conditional on the assumed start position.

2.7 Sequence checks and pipeline re-execution

The shared sequence checker separately evaluated a cloning-enzyme catalogue and a limited, sourced set of host restriction–modification motifs. It also screened promoter-like and Shine–Dalgarno-like patterns, long homopolymers and component junctions. These checks identify sequence features, not their activity. Baseline RNA-folding calculations used ViennaRNA [14]; edited P1 sequences were not assigned newly computed folding values.

Re-execution used Python 3.12.14, Biopython 1.85, HMMER 3.4 and ViennaRNA 2.7.2; remaining versions, commands, exit codes and logs are recorded in the audit evidence. Analyses ran in a separate working copy with the selected prior output files omitted before generation. The original archive was preserved. Fourteen third-party cached inputs excluded from the public deposit were restored from the supplied review archive only after their SHA-256 hashes matched `excluded_inputs.csv`; current downloads were not substituted. The public-only preparation step failed without the missing RegulonDB input. Source inspection identified the two RegulonDB records and three PRODORIC records as direct dependencies of the selected scan or follow-up stages. Regenerated files were compared with archived outputs by SHA-256 and byte equality. P1 generation consumed the freshly regenerated baseline.

The historical operator-scan runner selected `Ec_PhoP/Ec_OmpR` records, whereas its preparation function wrote `PhoP/OmpR` labels. Controlled execution of the original and corrected preparation functions tested the documented two-label mapping. The correction was already present in v0.6.0; no new scientific-code changes were required for the audit. These checks assess computational repeatability and specified numerical claims rather than biological model validity.

The published candidate audit was run on both the deposited collection and the freshly generated P1 collection. It checked identifier and sequence uniqueness, nucleotide alphabet, CSV–FASTA equality, lengths, GC content, component reconstruction and documented changes between the baseline and P1. A separate descriptive-check script recalculated panel counts, revision categories, selected

protein-identity and motif quantities, and the Table 2 file totals from regenerated outputs and source records; it did not import the project's analysis code.

3. Results

3.1 Accession reconciliation changed the sensor coordinate system

Cached PacA records differed at their N termini: UniProt Q6D515 contained 611 residues and RefSeq WP_043878053.1 contained 657. Residues 2–611 of the shorter record matched residues 48–657 of the longer one. The shorter record omitted the region containing the first predicted transmembrane helix. The sensor designs therefore used the longer RefSeq sequence and its coordinate system.

PhoQ also illustrated the distinction between sequence annotation and established protein function. Its HisKA profile score fell below the selected Pfam gathering threshold despite other evidence for its kinase function and catalytic histidine. The annotation retained the unresolved profile assignment rather than interpreting it as absence of the biological domain.

3.2 Protein comparisons distinguish sequence similarity from signaling specificity

The outputs contained 156 full-length comparisons, 52 REC-domain comparisons and 24 DHp comparisons (two donors against 12 native kinases with extractable DHp assignments). CitA (cg0089) had no detectable hit to the tested HisKA-family profiles and was excluded from the DHp identity matrix. For native MtrA, cg0862, REC aligned-column identity was 36.04% to *Salmonella* PhoP (40 identical positions in 111 columns) and 48.65% to *E. coli* OmpR (54 in 111). The 12.61-percentage-point difference was the largest absolute donor-identity gap among the 13 native regulators (Figure 1A), making MtrA the strongest contrast under this particular metric.

The full matrix supports inspection of that comparison but does not establish signaling orthogonality, meaning functional separation from the host network. Sequence identity alone does not measure phosphotransfer specificity, competition for phosphorylation, regulator–DNA interactions or expression-dependent effects.

3.3 Upstream motif counts depended strongly on background composition

The 3,282,708-base host genome had 53.84% GC, whereas 414,950 bases in 2,509 intergenic segments had 47.26% GC. Using intergenic background composition reduced the PhoP matrix information content from 9.56 to 8.28 bits. Of the original 304 flagged sites, 82 retained an empirical tail fraction $\leq 10^{-4}$ against the revised null (Figure 1B). Applying the nominal threshold to 535,264 eligible windows gives 53.5 expected exceedances; the shuffled realization gives a window-level reference count of 64. These cannot be compared directly with the 82 selected, merged sites to infer enrichment or a chance-level residual. Such an inference requires null replicates

processed through the same full selection procedure. The result demonstrates sensitivity to background choice rather than a measured change in binding.

The *E. coli* implementation check recovered 26 of 29 matrix-input sites at their exact coordinates. Because those sites trained the matrix, this is an implementation result rather than independent predictive accuracy. Peak merging also makes exact-coordinate recovery distinct from detection of a broader site region.

A predicted site upstream of *lysC* occupied plus-strand coordinates 269204–269220. The cached TSS dataset placed the associated start at 269335, giving a downstream-edge offset of –115 and a center at –123. That center is more distal than the proximal PhoP sites associated with Class I and Class II activation in *Salmonella*, where promoter architectures also vary [15]. Whether this site permits regulation at that distance in *C. glutamicum* remains unknown.

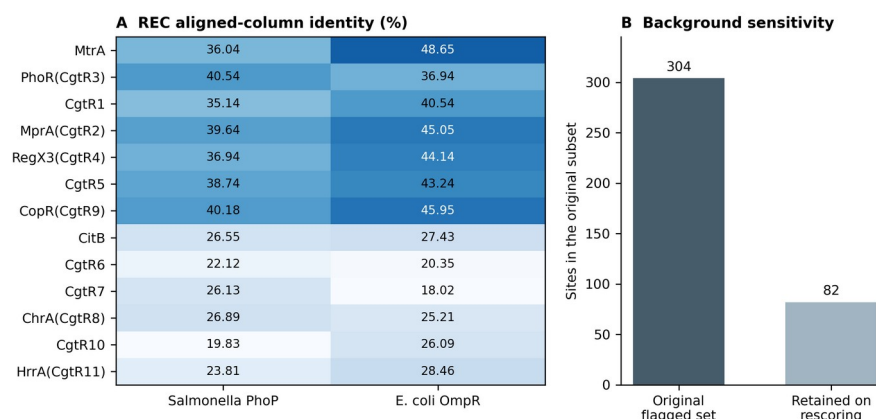


Figure 1. Comparative screens and their interpretation. A, REC-domain identity to the 13 native regulators in the supplied comparison set, using the aligned-column denominator. These percentages are not cross-talk probabilities. B, 82 of the original 304 upstream flagged sites retained the empirical threshold after intergenic-background rescoring. The right-hand bar is not a new whole-genome total.

3.4 The retained collection separates output comparisons from sensor alternatives

The collection separates 73 output-layer fragments from 24 sensor candidates: 18 PhoQ and six EnvZ (Table 1). Output families probe operator placement and sequence, promoter context and translation architecture. Sensor families sample junction alternatives around shared components. The collection is therefore a set of related comparisons rather than 97 independent complete circuits.

Table 1. Candidate families in the retained P1 collection.

Family	Count	Primary comparison
Operator substitution spacing	14	Position within a fixed-length background
Operator insertion spacing	18	Spacing with additional context and length changes
Background and spacer swaps	3	Local sequence context at fixed operator identity and spacing
Other promoter and operator comparisons	11	Core, operator and surrounding-sequence comparisons
Retained promoter context	4	Dependence on additional native/source context
Promoter–fore-cistron combinations	9	Interactions between output and translation parts
Translation tuning and mechanism	8	Fore-cistron selection and initiation-element perturbations
Leadered translation probes	2	Translation architecture in PsodA contexts
Calibration anchors	2	Source-context reference constructs
PacA–PhoQ sensor candidates	12	Four junction families with three local offsets each
McpX–PhoQ sensor candidates	6	Two junction families with three offsets each
Alternative EnvZ sensor candidates	6	Two junction families with three offsets each
Added matched operator controls	2	P1 parent context with a composition-matched scramble
Total	97	73 output-layer fragments and 24 sensor candidates

The 14-position substitution series spans 13 base-pair increments, approximately 1.24 DNA turns under a 10.5-base-pair-per-turn assumption. It jointly varies operator distance and rotational orientation. With a functioning output and adequate precision, response patterns could motivate tests of helical-phase dependence; absence of periodicity in an inactive or noisy system would be uninformative. Background and spacer swaps at fixed spacing probe local sequence effects. Separate operator variants and composition-matched scrambles probe operator-sequence dependence. Scrambling can retain partial motif features, so these are not established binding-null controls.

Translation-initiation and downstream-element perturbations provide additional comparisons of mechanisms that could yield similar reporter readings. The six EnvZ fragments preserve an alternative signaling branch, but a compatible OmpR output

remains necessary. Candidate-level rationales, component origins and sequences accompany the supplementary catalogue.

3.5 Two added controls expand the baseline from 95 to 97 fragments

The baseline already contained two-base TC insertions in 17 Psyn fore-cistron designs. P1 separately extended 39 plain Psyn output-layer fragments by two bases under a direct downstream start-codon assumption and added a 13-base translation-initiation element to two PsodA probes. The remaining 54 baseline records were unchanged. Adding two matched operator-scramble controls then produced the 97-record collection (Figure 2).

The new controls preserve their P1 parents outside the annotated 16-base operator and retain nucleotide composition within it. Direct sequence checks confirmed these properties. The 24 sensor fragments remained at version 4; the revision affected the output layer.

These edits express conditional design choices. The source publications do not establish the TSS at the new reporter junction, and the context-sensitive model cannot resolve it experimentally. Changes in the eventual vector-reporter assembly could therefore require reassessment; the P1 edits are not demonstrated corrections of a failed biological construct.

3.6 Pipeline re-execution recovered the selected outputs

The additional audit reproduced all 38 selected files byte for byte across eight analysis or generation stages (Table 2), including the five representative outputs specified in the author-verification checklist. The targets covered the baseline manifest, genome-scan output, sensor sequences, REC-domain identity comparison and P1 FASTA. One additional background-summary JSON, outside the selected 38, differed in two historical Poisson values, each by less than 10^{-16} . The difference was retained as a failure of strict byte equality; the site counts and selected outputs were unchanged. The comparison excludes refetching historical source data and rerunning DeepTMHMM inference.

Table 2. Selected files reproduced byte for byte.

Stage	Files	Scope
Protein comparison	7	Identity tables and extracted-domain records
Domain annotation	2	Tables regenerated using cached topology inference
Sensor generation	2	Version 4 CSV and FASTA
Transcript-start analysis	2	Distribution and historical assessment report
Operator scan	4	Correction already included in v0.6.0
Follow-up analysis	7	Background/site results and associated records
Baseline consolidation	8	95-fragment manifest and companion outputs

Stage	Files	Scope
P1 generation	6	97-fragment proposal and edit/check outputs
Total	38	All files listed here are byte-identical

Controlled testing reproduced the historical runner failure: the original labels selected zero PhoP control records and led to an `IndexError`, whereas the corrected labels selected 29. Across the 81 prepared site rows, only the regulator-label field changed; all other fields matched, and the site FASTAs and PWM files were byte-identical between versions. The corrected full pipeline completed. Thus, the result supports restoration of record selection without changes to the prepared sequences, coordinates or PWM values; it does not imply that the unsuccessful original runner produced equivalent final outputs. The audit comparison tables preserve the selected paths, archived and regenerated hashes, failure logs and precision exception.

The sequence-check, register-check and numerical-report suites passed 41, 23 and 47 checks, respectively. The separate P1 audit found 97 unique identifiers and nucleotide strings, matching CSV and FASTA sequences, and exact component reconstruction. Lengths ranged from 66 to 1,842 bases and GC content from 31.00% to 53.83%. The additional descriptive-check script passed 64 checks, including every family count in Table 1, the 95-to-97 reconciliation and the stage totals in Table 2; file, row-selection and calculation details are retained in the audit evidence. Baseline consolidation reproduced its folding-derived fields; folding was not recomputed for the edited P1 sequences.

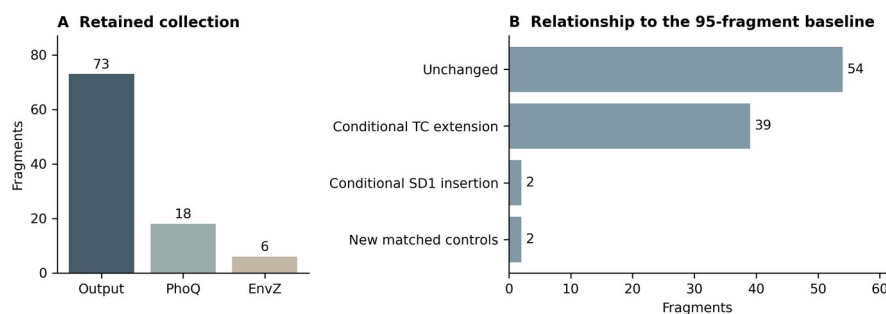


Figure 2. Composition and revision history of the P1 collection. A, retained output and sensor fragment counts. B, relation to the 95-fragment baseline. The 39 conditional TC extensions are separate from the 17 fore-cistron interface edits already present in the baseline. Counts describe fragments, not complete circuits or independent biological trials.

4. Discussion

The workflow exposed dependencies that changed the interpretation of the design: source-record choice altered sensor coordinates, background composition altered

motif screening, and reporter-junction assumptions altered the translation architecture. These findings make the candidate collection more precisely specified and identify which assumptions a subsequent experiment would test.

The transferable contribution is the link between design choices, explicit component intervals and regenerable outputs. Recorded denominators, version boundaries and source corrections make numerical and sequence claims inspectable. Provenance and machine-readable outputs support the aims of the FAIR Guiding Principles [16], and the public release provides a persistent DOI and stated reuse terms. However, excluded third-party inputs limit complete regeneration from the public deposit alone. Retaining commands, software versions and intermediate results follows recommendations for reproducible computational research [17].

Biological compatibility remains unresolved. Ligand binding must influence the fused kinase, the kinase must regulate its response regulator, and that regulator must control the designed promoter in *C. glutamicum*. Reporter behavior could also change through translation, growth or betaine-dependent physiology. Shared dependencies prevent treating sequence diversity as independent chances of success. In a related cautionary example, Li et al. observed reduced reporter expression after inducer addition for their reconstructed BetI system, PbetI2R1, in *C. glutamicum* and the two other tested hosts [18]. This establishes an undesired response direction for that system, not failure of the betaine-responsive kinase architecture proposed here.

Computational limits also constrain interpretation. The host-genome scan uses a small *E. coli*-derived motif set for a *Salmonella* regulator, with *Salmonella* sites providing a comparison rather than host-specific occupancy evidence. Overlapping windows and limited null sampling restrict significance claims. The transcript-start benchmark is anchored to known starts, so its apparent accuracy cannot be extrapolated to untested promoter junctions or justify the probability language retained in some internal design analyses.

The resource provides candidate fragments rather than a finished experimental build. Full plasmid contexts, regulator-expression arrangements and a compatible alternative-branch output remain to be specified. No fluorescence, growth, dose-response, transcript or production measurements were generated, and neither useful regulation nor an industrial benefit has been demonstrated.

The collection allows other groups to inspect, modify or test a defined design without reconstructing its provenance. A useful first experimental priority would be to assess the output layer with PhoP activation independently established in *C. glutamicum* before introducing chimeric-sensor uncertainty. Native PhoQ/PhoP offers one possible route: PhoQ responds to Mg^{2+} depletion in *Salmonella* [19], but efficient transduction of that signal in *C. glutamicum* would itself need verification.

Data and code availability

The candidate CSV and FASTA collection, component maps, baseline records, candidate rationales, analysis scripts, available cached inputs, original re-execution

records, label-correction patch and figure code are available at <https://github.com/gabeykim/cglut-betaine-circuit>. The audited source is release v0.6.0, archived at <https://zenodo.org/records/23132170> (DOI: 10.5281/zenodo.23132170). Fourteen third-party cached files are excluded from the public deposit; their source locations and SHA-256 hashes are recorded in `excluded_inputs.csv` and the source notices. Exact copies from the supplied review archive enabled the additional audit. Its report, commands, logs, numerical-claim table and regenerated-output comparisons accompany this manuscript revision as `Glutamicum_Reproducibility_Audit.pdf` and `Glutamicum_Reproducibility_Audit_Evidence.zip`; these additional records are not part of frozen release v0.6.0. Historical reports are retained for provenance, with the corrections and interpretive limits stated here taking precedence.

Original project code is released under the MIT License. The manuscript and original sequence-design and analysis data are released under the Creative Commons Attribution 4.0 International License (CC BY 4.0), to the extent the author holds applicable rights. Third-party assets retain their original rights and attribution requirements.

Author statements

Author contributions (CRediT): Gabriel Kim — Conceptualization, Methodology, Formal analysis, Project administration, Writing – review & editing. Software development, computational analysis and initial drafting used the tools described in the AI disclosure. Affiliation: Stanford University, Stanford, CA, USA. Correspondence: gabeykim@stanford.edu.

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AI disclosure: Claude (Anthropic) was used for design reasoning, coding and project documentation. ChatGPT (OpenAI) was used for code review and correction, command execution, output comparisons, literature checking, figures and manuscript preparation. For the additional audit, ChatGPT executed the pipeline and controlled runner comparison, implemented the descriptive checks, compared hashes and prepared the audit report. Exact model variants used in the earlier work are not established by the retained records. Responsibility for the final analyses and claims rests with the human author.

Verification record: The additional audit was completed on 4 October 2026. Its commands, environment, file identities, expected and observed hashes, numerical checks and limitations are provided in the accompanying audit evidence. The author subsequently reran the five-output verification checklist and reviewed the resulting records.

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